Another issue with simulated annealing is that tuning its parameters can be tricky and annoying. In **adaptive simulated annealing** the algorithm parameters, k and α, are automatically adjusted as the algorithm progresses, or depending on trend in the energy level; the latter uses mechanisms borrowed from thermodynamics, allowing even increases in temperature (simulating cycles of cooling down and warming up).

17.1.6 Simulated Annealing vs Gradient Descent: Which One Should I Use?

We have seen that simulated annealing is slower than gradient descent to get to minima: the latter takes the fastest route, so it's hard to beat on a clear course.

The caveat is that gradient descent requires a differentiable cost function, and gets easily stuck in local minima.

Whenever the cost function is not differentiable or step-shaped, like for the minimum-crossing-embedding problem, or the problem space is discrete (for problems like the TSP, which we'll describe later in this chapter), then simulated annealing is preferred over gradient descent.

Likewise, for problems where we have flexible requirements about running time and the quality of the solution, simulated annealing might still be preferable to gradient descent: keep in mind that simulated annealing is a *Monte Carlo algorithm*, and so (as we discuss in appendix F) it returns a sub-optimal